

List of parameters:

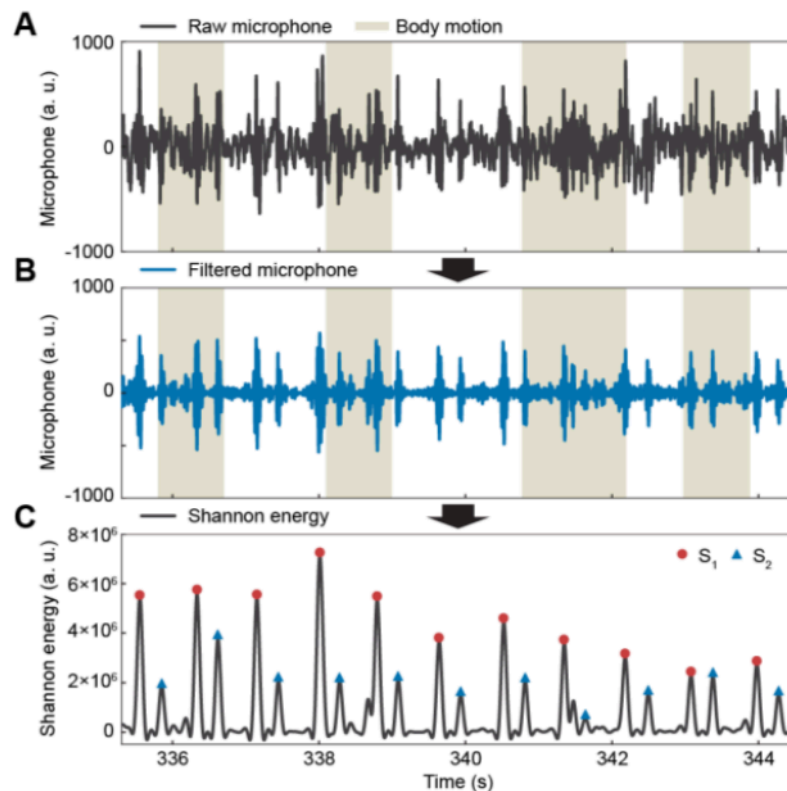
- CSI (cardiac sound intensity)
- HR
- HRV (RMSSD)
- HRV (LF/HF)
- HRV (LF) and HRV (HF)
- RR
- RRV

1) CSI

- Step 1: Bandpass filter heart sound: 20-200 Hz
- Step 2: Shannon energy calculated: $S = C \times \log(C)$

where S is the Shannon energy and C is the cardiac sound, to yield clear peaks associated with cardiac cycles

- Step 3: Detect S_1 and S_2 peaks in Shannon energy. The peaks of the Shannon energy envelope defined the timing of the S_1 and S_2 features. The amplitude ratio of S_2/S_1 served as the basis for CSI $\rightarrow \text{CSI} = S_2/S_1$



- CSI were continuously monitored every second by averaging the intervals and sound intensities of the 10 most recent cardiac sound peaks.

2) HR

- Step 1: Low pass filter ECG signal 150 Hz, detrend (DC offset remove)
- Step 2: Define negative peaks (Pan–Tompkins, ...)

- Step 3: Calculate the interval between peaks, called NN series (NN mean normal to normal, in some paper, it write RR series as R peak to R peak, but in here, I call NN), NN in “ms” unit
- Step 4: Reject any NN interval shorter than 300 ms (200 bpm) or longer than 1500 ms (40 bpm). A very reliable standard is to reject any NN interval that differs from the median of the surrounding 5–10 beats by more than **20%**.
- Step 5: Replace the rejected intervals using **cubic spline interpolation** or simple linear interpolation using the surrounding valid NN intervals.
- Step 6: Calculate HR from NN:
HR = 60 / NN

3) HRV (RMSSD)

- From the NN series in above (after reject bad NN and interpolate NN), calculate HRV (RMSSD)

The Mathematical Formula

The formula breaks down exactly what the name implies, working from right to left:

$$RMSSD = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N-1} (NN_{i+1} - NN_i)^2}$$

Where:

- NN_i is the duration of the current interval (usually in milliseconds).
- NN_{i+1} is the duration of the very next interval.
- $(NN_{i+1} - NN_i)$ is the **successive difference** between consecutive beats.
- N is the total number of NN intervals in your time window.

4) HRV (LF), HRV (HF)

- Step 1: From the NN series above (after rejecting bad NN and interpolate NN), apply the **Welch method** or a standard FFT with a Hanning window to compute the power spectrum. Note: window size at least 1 min
- Step 2: Calculate the Area Under the Curve (AUC) within the specific frequency brackets: integrate from 0.04 to 0.15 Hz for **LF power**, and from 0.15 to 0.40 Hz for **HF power** (expressed in ms²)

5) HRV (LF/HF)

- Take LF and HF from above, LF/HF is ratio between them

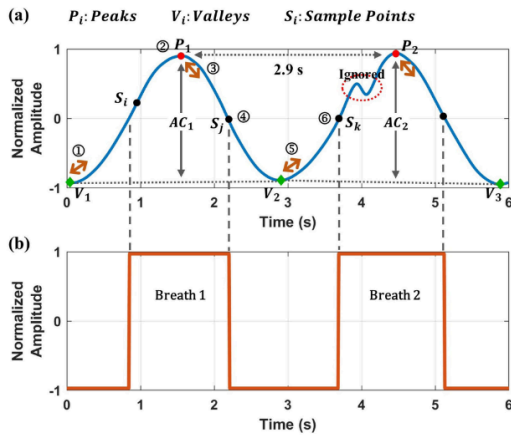
6) RR

- Step 1: Detrend (DC offset remove)
- Step 2: A moving average filter with a sliding window of 0.5 s is used to smooth the raw BR signal
- Step 3: Apply fourth-order low-pass Chebyshev II filter with a stopband edge frequency of 1 Hz, followed by a second-order high-pass Butterworth filter with a cut-off frequency of 0.12 Hz. (As the frequency of BR is very small, if we apply normal filter, the data can be distorted → need high order filter, more sharp filter)

- Step 4: Feature point detection (copy from paper)

The peaks and troughs of the BR are determined to calculate the RR. The peak point is a local maximum, which occurs when the signal changes from a rising slope to a falling slope. Identifying the rising slope is necessary before locating the peak point. We empirically set the rising slope to be at least 250 ms of continuous increase, and the falling slope to be at least 250 ms of continuous drop. At a sampling frequency of 500 Hz, this window has 125 sample points. Setting a window period of 250 ms can help ignore ripples and fluctuations caused by body movement noise.

After identifying a rising slope (step 1a), each new sample point (S_i) is compared to a temporary local maximum value (P_{tmp}), and P_{tmp} is constantly updated until a local peak is reached (step 2a) and the falling slope occurs (step 3a). The peak (P_1) between the rising slope and the falling slope is temporarily defined as the peak point of a respiratory cycle. The point (P_1) can be confirmed as the peak of a respiratory cycle if the waveform continues to decrease and the fall exceeds half of the previous peak, that is, when the sample point $S_j \leq 0.5 \times (V_1 + P_1)$, as shown in step 4a. Similarly, if the rising slope is detected (step 5a), and the latest sample point S_k on the rising slope is greater than half of the previous peak, that is, $S_k \geq 0.5 \times (V_2 + P_1)$, the point previously defined as the local minima (V_2) can be confirmed as the onset of the respiratory cycle. In this way, the feature points of a breathing cycle can be detected.



- Step 5: RR calculation (copy from paper)

When the peak point and two side troughs are obtained, the amplitude of signal is the distance between P_1 to the line of $V_1 V_2$ (For example, P_1, V_1, V_2 is 1 triangle, then amplitude of signal is the height from P_1 down to the line between $V_1 V_2$). After that, the amplitude of each cycle (AC_i) is compared to an adaptive threshold to verify that an effective respiratory cycle has been identified. The adaptive threshold is adjusted automatically using the formula below from the last eight valid respiratory amplitudes. The threshold is empirically established at one-third of the average amplitude to adapt to variations in the subject's breathing intensity. When an effective respiratory cycle is found, the RR can be calculated on the basis of the time interval between two successive breathing peaks.

$$\text{Adaptive_Threshold} = \frac{1}{3} \cdot \left(\frac{1}{8} \cdot \sum_{i=n-7}^n AC_i \right) \quad (1)$$

7) RRV

The values of RR and RRV followed from averaging the intervals between the five most recent respiratory peaks and calculating the deviations in these intervals, respectively.